THE BLACK-SCHOLES MODEL

The Black–Scholes model is one of the most influential equations in mathematical finance. The underlying framework that the Black-Scholes model provides is rich in foundational ideas for determining the fair value of a European option.

3.1 Assumptions

Modelling Asset Prices with Stochastic Processes: The underlying asset, S, follows a geometric Brownian motion (GBM) (lognormal random walk of the form):

$$\frac{dS}{S} = \mu dt + \sigma dX, \tag{1}$$

where μ is the long-term growth rate of S, σ is the volatility and X is a R.V. given by $dX = \phi(dt)^{\frac{1}{2}}$ where $\phi \sim N(0,1)$ (*i.e.,* follows a standard normal distribution).

No-Arbitrage and Replication: The Black-Scholes equation is derived under the assumption of Delta hedging by considering a portfolio of the form $\pi = V - \Delta S$, where the holder of the portfolio continuously buys or sells shares in S so that Δ is instantaneously equal to $\frac{\partial V}{\partial S}$. This portfolio is then risk-free with $\frac{d\pi}{dt} = r\pi$. Hedging is the idea of taking an opposing position for the existing underlying asset, to reduce or offset risk. Arbitrage refers to a situation where an investor can make an instantaneous risk-free profit without the probability of loss. Whilst no-arbitrage constraints are the conditions imposed on a model so that no risk-free profit opportunities exist.

Parameter Constants: The parameters (μ, σ, r) are all assumed to be constant, where r is taken to be the risk-free rate.

3.2 Fair Value of a European Call Option

In the Black–Scholes model, the fair price of a European call option is given by:

$$C = Se^{-qT}\Phi\left(d_1\right) - Ee^{-rT}\Phi\left(d_2\right), \tag{2}$$

$$d_1 = \frac{\ln\left(\frac{S}{E}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T},$$

where $\Phi(x)$ denotes the cumulative distribution function (CDF) of the standard normal distribution. At a high level, (2) is the weighted difference between the asset and strike prices.

3.3 Intuition of the Black-Scholes Model

If asset prices evolve randomly in a lognormal way but consistent with constant interest rates and volatility, we can ask the question: what must an option be worth at any time, t, for a perfectly hedged position to just break even? The Black-Scholes model is derived under the assumption of Delta hedging, using the hedge ratio, Δ (number of shares). In a perfectly hedged position, the change in the option's value and the change in the value of the hedge cancel out, so the combined position has zero instantaneous PnL (where PnL stands for Profit and Loss), it does not make nor lose money. In other words, $\text{Option}_{PnL} + \text{Shares}_{PnL} = 0$, which ensures that the hedged portfolio earns the risk-free rate, and this no-arbitrage requirement uniquely determines the option's fair price. The Black-Scholes equation is:

$$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + (r - q)\, S\frac{\partial C}{\partial S} - rC = 0. \tag{3}$$

This is the relationship between how two key variables, the underlying asset price, S, and time, t, will affect the value of a European call option, C. The Black-Scholes model is a perfect introduction to the Greeks, which are sensitivities of option prices to underlying parameters which are used for hedging:

$$\Delta = \frac{\partial C}{\partial S}, \qquad \Gamma = \frac{\partial^2 C}{\partial S^2} = \frac{\partial \Delta}{\partial S}, \qquad \Theta = \frac{\partial C}{\partial t}. \tag{4}$$

Delta, Δ, is how the call price will change with a change in the asset price. Gamma, Γ, is the rate of change of Δ. And Theta, Θ, is how the call price will change as time passes. Thus, we obtain:

$$\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma + (r - q)\, S\Delta - rC = 0. \tag{5}$$

There are other terms here, such as $(r-q)\,S$ which represents the forward price of the asset (*i.e.*, the value of the asset at expiration if only the interest affects the price) and rC which represents the call option's value back to present (*i.e.*, removes the effect of interest from the option's value at expiration). Finally, σ, which is the standard deviation and determines the width of the assumed lognormal distribution.

To calculate effects, like the Greeks and the call price, the Black-Scholes equation must be solved. Let us further analyse and see what each of the terms in (2) really mean. The term $S\Phi\left(d_1\right)$ denotes that if I held an option until expiration, what would the average value of the asset be above the strike price, *i.e.*, it denotes the amount you will earn for owning this option on average (which we can denote by Profit_{Avg}). The e^{-qt} term means the price is now the dividend-adjusted price. The term $e^{-rT}\Phi(d_2)$ denotes what is the likelihood of paying the exercise price at expiration, *i.e.*, the likelihood of finishing in the money. Multiplying by the strike price, E, we can see how much it will cost on average to own this option (which we can denote by Cost_{Avg}). Thus:

$$C = \text{Profit}_{Avg} - \text{Cost}_{Avg}. \tag{6}$$

These terms represent the current expected value of the option by subtracting the average cost from the average profit we will earn from the option, so we can work out its fair value.

3.4 Geometric Brownian Motion Implies Lognormality

In the assumptions of the Black-Scholes model, the underlying asset, S, follows a GBM (1). This describes how the asset price will fluctuate with time (*i.e.*, the path). This is equivalent to saying that the underlying asset, S, follows a lognormal random walk. When you solve the GBM equation, you find that the asset price can be written as an exponential of a normally distributed term (with that term being $dX = \phi(dt)^{\frac{1}{2}}$). In other words, the logarithm of the price is normally distributed, which means that the underlying asset follows a lognormal distribution. Thus, the lognormal distribution is not a separate assumption, rather it is a mathematical consequence of the underlying asset, S, following a GBM.

3.5 Adjusting to the Lognormal from the Normal Distribution

The Black-Scholes model follows a lognormal random walk, to adjust to a lognormal distribution we just take the natural log ($\ln$) of the values. Thus, we transform the standard normal distribution:

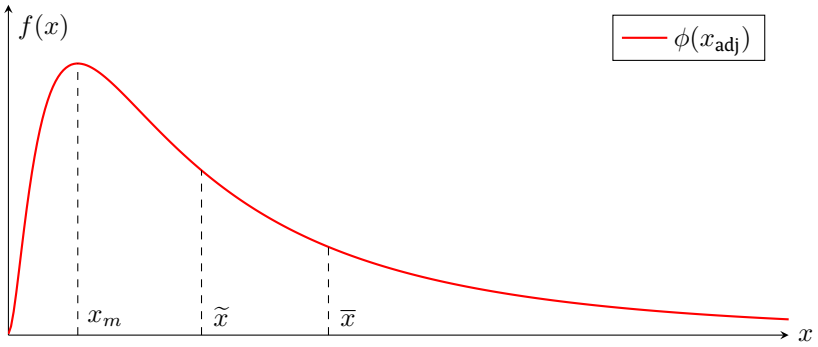


Figure 1: Lognormal Distribution

The mode, x_m, is the peak of the distribution curve and is the value at which occurrences happen most frequently. The median, $\widetilde{x}$, is the midpoint, where half the occurrences lie to the left and half to the right. The mean, $\overline{x}$, is the average of the distribution, where the total value is split equally. In a standard normal distribution, these three points coincide, but in a lognormal distribution they are separate, reflecting its skewed shape.

In a normal distribution, the relationship between the asset price and strike price is captured by taking the difference between them, $(S-E)$. However, in a lognormal distribution, we must compare the natural logarithms of both, giving $\ln(S) - \ln(E) = \ln\left(\frac{S}{E}\right)$. Thus, if $S > E$, we obtain $\ln\left(\frac{S}{E}\right) > 0$ and so, the call is in the money. If $S < E$, then we obtain $\ln\left(\frac{S}{E}\right) < 0$ and so, the call is out of the money.

We also need to adjust for other components that significantly affect option pricing: interest rates and volatility. Since options are based off the forward price of the underlying, and the forward price depends on interest rates, we need to account for interest rates over the life of the option, $(r-q)T$. This ensures pricing reflects the present value of the asset's expected price at expiration. In a lognormal distribution, asset prices are right-skewed, which results in the mean being shifted slightly to the right of the mode, as seen in Figure

(1). Mathematically, this equals $\frac{1}{2}\sigma^2 T$. Together, these adjustments represent the numerator for d_1 in the Black-Scholes model:

$$\begin{aligned} d_1^{\text{num}} &= \ln\left(\frac{S}{E}\right) + (r-q)T + \frac{1}{2}\sigma^2 T \\ &= \ln\left(\frac{S}{E}\right) + \left(r - q + \frac{1}{2}\sigma^2\right) T. \end{aligned}$$

To find d_1, we must convert into a number of standard deviations because standard deviations measure how far the strike price is from the mean in a lognormal distribution. Over any time interval, T, one standard deviation is equal to $\sigma\sqrt{T}$. The square root appears because volatility increases with time but not linearly, price dispersion widens at a slower rate as time passes. Therefore:

$$d_1 = \frac{\ln\left(\frac{S}{E}\right) + \left(r - q + \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}}, \tag{7}$$

d_1 determines how many standard deviations the strike price is from the mean. Now, combining with the forward price of the asset gives as the average profit $\left(\text{Profit}_T = Se^{rT}\Phi(d_1)\right)$ at expiration. To figure out the likelihood that the option will be exercised, we can adjust d_1 to use the median from the mean in a lognormal distribution, this difference is given by $\sigma\sqrt{T}$, and so, $d_2 = d_1 - \sigma\sqrt{T}$. The term $\Phi(d_2)$ uses the median to estimate the probability of the option being in the money at expiration. Now, multiplying this probability by the exercise price gives the average price paid at expiration $(\text{Cost}_T = E\Phi(d_2))$. Thus, combining with the average profit at expiration, we obtain the expected value of the call option at expiration, C_T:

$$\begin{aligned} C_T &= \text{Profit}_T - \text{Cost}_T \\ &= Se^{rT}\Phi(d_1) - E\Phi(d_2). \end{aligned} \tag{8}$$

Since, we pay for the option at the present value, we must adjust by multiplying both terms by e^{-rT}, and adjust the price to the dividend-adjusted price. Thus, we obtain the fair value of a European call option for the Black-Scholes model:

$$C = Se^{-qT}\Phi\left(d_1\right) - Ee^{-rT}\Phi\left(d_2\right). \tag{9}$$

It should be known that the Black-Scholes model is a complete option-pricing model, Dupire's model is fundamentally a volatility model that becomes an option pricing model when incorporated with a stochastic differential equation (SDE), Heston's model is a volatility model and an option pricing model since it gives the stochastic process for volatility and asset price. A risk-neutral diffusion is simply the SDE that describes how an asset evolves under the risk-neutral measure, the probability measure used for pricing derivatives. Under this measure, all tradable assets must grow at the risk-free rate, and the discounted asset price must be a martingale (a martingale condition is a stochastic process, a sequence of random variables, where the conditional expectation of the next value, given all prior observations, is equal to the current value). For the Black-Scholes model, Dupire local volatility model and Heston stochastic volatility model, the risk-neutral diffusion is a GBM with constant volatility, GBM with local volatility, and two-factor system with stochastic variance and asset price, respectively.

3.6 Newton-Raphson Method to Find Implied Volatility

From the data, we have five parameters which are known and one which is not, σ. Thus, we want to minimise the difference between the theoretical price (which is the price from the Black-Scholes function) and the market price (the mid price calculated from bid-ask spreads in the option chain). We can do this by the Newton-Raphson method. If we denote x^g as the initial guess for the volatility. Then, by the Newton-Raphson method, we can iterate to a closer value for volatility to the true value:

$$x_{n+1}^g = x_n^g - \frac{f\left(x_n^g\right)}{f'\left(x_n^g\right)}. \tag{10}$$

Here, $f\left(x_n^g\right)$ is the difference between the theoretical price (P_{theory}, the Black-Scholes valuation using the initial guess for the volatility) and the actual price (P_{actual}, the mid price). $f'\left(x_n^g\right)$ is the gradient of the difference. More explicitly, it is the partial derivative of the difference with respect to the volatility. If we denote the difference as $\mathcal{L}$, then:

$$f'\left(x_n^g\right) = \frac{\partial \mathcal{L}}{\partial \sigma}. \tag{11}$$

Thus, we can relabel the Newton-Raphson formula:

$$\sigma_{n+1} = \sigma_n - \frac{\mathcal{L}(\sigma_n)}{\frac{\partial \mathcal{L}}{\partial \sigma}}. \tag{12}$$

Thus, we just solve for implied volatility iteratively:

Step 1. Make a guess for volatility.
Step 2. Calculate the difference, $\mathcal{L}$.
Step 3. Check if the difference is less than the tolerance, ϵ: $|P_{theory} - P_{actual}| < \epsilon$.
Step 4. Calculate the gradient of the difference, $\frac{\partial \mathcal{L}}{\partial \sigma}$.
Step 5. Update the volatility using the Newton-Raphson formula.
Step 6. Repeat until converged or maximum number of iterations, N_{iter}.

To enhance the speed, or to ensure convergence, there are a number of methods to choose an initial value for σ. If options are near at the money, then we can use the Brenner-Subrahmanyam approximation, which approximates the Black-Scholes price by assuming zero interest rates and dividends and using a Taylor series expansion:

$$x^g = \sqrt{\frac{2\pi}{T}} \frac{C}{S}. \tag{13}$$

Alternatively, we can use a constant (*e.g.*, $x^g = 0.8$) or a simple heuristic approach using log moneyness (*e.g.*, $= 0.2 + 0.1 \ln\left(\frac{S}{E}\right)$).

3.7 Plotting the Implied Volatility Surface

We can find the implied volatility for all strikes and maturities. In this case, I have chosen the x-axis to be 'Moneyness' $\left(= \frac{S}{E}\right)$ and the y-axis to be 'Time to Expiration'.

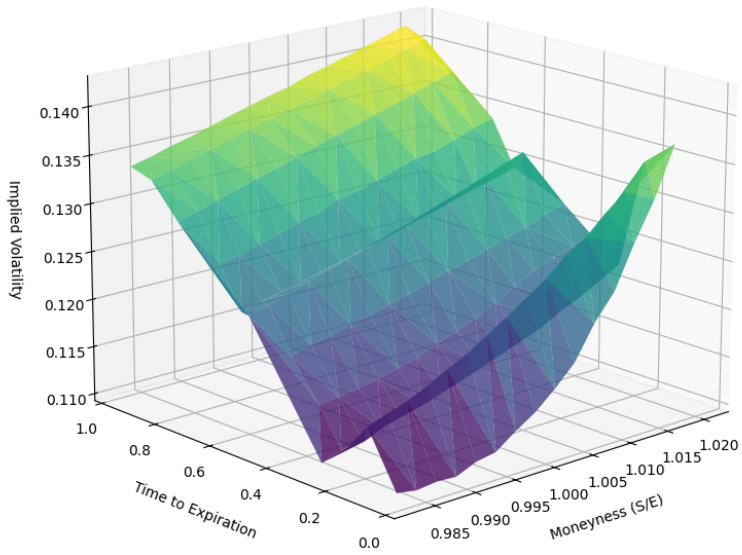


Figure 2: Implied Volatility Surface.

Figure (2) displays the Black-Scholes output using Python code for the dataset.